

Equity Financing Restraint and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Financing Restraint (EFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EFR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.93 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, yet the cross-sectional variation in expected returns remains only partially explained by traditional consumption-based models. We examine how firms’ equity financing decisions relate to their exposure to systematic consumption risk, focusing on a novel measure we term Equity Financing Restraint (EFR). This signal captures firms’ reluctance to issue new equity despite potentially profitable investment opportunities, suggesting these firms face higher costs of external financing during periods when investors’ marginal utility of consumption is high. The consumption-based asset pricing framework predicts that firms exhibiting greater financing restraint should command higher risk premia if their cash flows covary more strongly with aggregate consumption growth or if they are particularly vulnerable during bad economic states when the representative investor’s marginal utility is elevated.

The theoretical link between equity financing restraint and expected returns emerges through firms’ differential exposure to consumption risk across economic states. Following the long-run risks model of [Bansal and Yaron \(2004\)](#), firms that restrict equity issuance likely face binding constraints precisely when consumption growth is low and uncertain, amplifying their systematic risk exposure. These firms cannot easily smooth cash flows through external financing when the stochastic discount factor is high, making their equity claims particularly risky from a consumption-hedging perspective. The habit formation framework of [Campbell and Cochrane \(1999\)](#) provides additional theoretical support: firms with high EFR exhibit greater sensitivity to surplus consumption ratios, as their inability to raise capital during economic downturns—when habit levels bind—exposes investors to heightened consumption risk.

The state-dependent nature of financing constraints connects directly to the in-

tertemporal marginal rate of substitution through disaster risk considerations. Building on [Barro \(2006\)](#) and [Gabaix \(2012\)](#), firms that voluntarily restrict equity financing signal their vulnerability to rare consumption disasters, as maintaining financial flexibility becomes prohibitively expensive when disaster probabilities increase. This mechanism implies that high-EFR firms load positively on consumption-based factors that capture tail risk exposure, consistent with the variable rare disasters framework where time-varying disaster probabilities drive both consumption dynamics and asset prices.

The consumption-based interpretation of EFR also relates to the broader literature on financial constraints and systematic risk. [Livdan et al. \(2009\)](#) demonstrate theoretically that financially constrained firms exhibit higher sensitivity to aggregate productivity shocks that drive consumption growth, while [Gomes and Schmid \(2010\)](#) show that financing frictions amplify firms' exposure to systematic risk through their effect on investment timing. Our EFR measure captures this exposure by identifying firms whose financing decisions reveal their heightened sensitivity to consumption risk, particularly during periods of elevated marginal utility when external finance becomes most costly relative to the stochastic discount factor.

We found that a value-weighted long/short trading strategy based on EFR achieved an annualized Sharpe ratio of 0.53 for gross returns and 0.47 after accounting for transaction costs, demonstrating robust performance that places it in the top 8

The consumption risk interpretation gained support from the strategy's factor loadings, particularly its significant positive loading of 0.34 (t-statistic of 6.65) on the CMA factor, which captures firms' investment behavior and relates to their sensitivity to consumption growth dynamics. The strategy's performance proved robust across different portfolio construction methodologies and size segments, with the long/short EFR strategy among the largest quintile of stocks generating 24 basis

points per month (t-statistic of 2.68), confirming that the effect was not confined to small, illiquid securities where consumption risk effects might be confounded by market frictions.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EFR strategy maintained an alpha of 15 basis points per month with a t-statistic of 2.24, demonstrating that it captured unique information about firms' consumption risk exposure beyond existing predictors. The net generalized alphas remained positive and significant across all factor models after accounting for transaction costs, with the strategy improving the mean-variance efficient frontier in all cases tested, suggesting that EFR provides economically meaningful improvements to investors' opportunity sets consistent with compensation for bearing systematic consumption risk.

Our study extends the consumption-based asset pricing literature by introducing a novel measure that captures firms' exposure to consumption risk through their equity financing decisions, contributing to recent work examining the asset pricing implications of corporate policies. While [Pontiff and Woodgate \(2008\)](#) document that share issuance predicts returns and [Cooper et al. \(2008\)](#) show similar effects for asset growth in the *Journal of Finance*, our EFR measure provides distinct information by focusing on financing restraint rather than observed issuance activity, thereby capturing firms' forward-looking assessment of their consumption risk exposure. Unlike [Lyandres and Palazzo \(2019\)](#) who examine payout policy in the *Journal of Financial Economics*, we identify firms that simultaneously restrict both equity issuance and payouts, revealing a more comprehensive picture of financial constraints that relate to marginal utility dynamics.

Methodologically, we advance the literature by demonstrating that EFR maintains significant predictive power even after controlling for closely related anomalies including share issuance, asset growth, and payout measures, suggesting it captures

a unique dimension of consumption-based risk not reflected in existing factors. Our comprehensive testing framework, following [Novy-Marx and Velikov \(2023\)](#), provides robust evidence that EFR’s performance cannot be attributed to small-stock effects or transaction costs, addressing concerns raised in the Review of Financial Studies about the implementability of many documented anomalies. The strategy’s ability to expand the mean-variance efficient frontier net of costs across multiple factor models demonstrates its economic significance for practical portfolio management.

The broader implications of our findings suggest that corporate financing decisions contain valuable information about firms’ systematic risk exposures that traditional factor models fail to capture fully. By linking equity financing restraint to consumption-based pricing kernels, we provide a theoretical foundation for understanding why certain corporate policies predict returns, contributing to the ongoing debate about rational versus behavioral explanations for cross-sectional return predictability. Our results support the view that at least some portion of the documented return predictability in equity markets reflects compensation for bearing systematic consumption risk, particularly during states of the world when marginal utility is high and financing frictions bind most severely.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique capital structures and regulatory environments. Equity Financing Restraint signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents

the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the cumulative amount of equity capital raised through common stock issuances net of retirements. Invested capital (ICAPT) measures the total capital invested in the firm’s operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable provides a comprehensive measure of the permanent capital employed in the business. We construct the Equity Financing Restraint signal by calculating the year-over-year change in common stock scaled by lagged invested capital: $(CSTK(t) - CSTK(t-1)) / ICAPT(t-1)$. This ratio captures the extent of new equity financing activity relative to the firm’s existing capital base. A higher value indicates more aggressive equity issuance, while lower or negative values suggest restraint in accessing equity markets through either limited issuance activity or share repurchases. We compute this measure annually using fiscal year-end values and require non-missing observations for both CSTK and ICAPT in consecutive years. To ensure meaningful scaling, we exclude observations where lagged invested capital is zero or negative, and we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFR signal. Panel A plots the time-series of the mean, median, and interquartile range for EFR. On average, the cross-sectional mean (median) EFR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFR data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFR signal for the CRSP universe. On average, the EFR signal is available for 6.37% of CRSP names, which on average make up 7.68% of total

market capitalization.

4 Does EFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFR portfolio and sells the low EFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EFR strategy earns an average return of 0.31% per month with a t-statistic of 4.15. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.65 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 573 stocks and an average market capitalization of at least \$1,481 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.14. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 20-34bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.66. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the EFR strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFR, as well as average returns and alphas for long/short trading EFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EFR strategy achieves an average return of 24 bps/month with a t-statistic of 2.68. Among these large cap stocks, the alphas for the EFR strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.93 and 2.73.

5 How does EFR perform relative to the zoo?

Figure 2 puts the performance of EFR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EFR strategy falls in the distribution. The EFR strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFR strategy (red line).² Ignoring trading costs, a \$1 invested in the EFR strategy would have yielded \$7.45 which ranks the EFR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFR strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$5.41 which ranks the EFR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFR relative to those. Panel A shows that the EFR strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFR ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFR or at least to weaken the power EFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR}EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFR signal in these Fama-MacBeth regressions exceed 2.83, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFR is 1.47.

Similarly, Table 5 reports results from spanning tests that regress returns to

an average month.

the EFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFR strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFR trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does EFR add relative to the whole zoo?

Finally, we can ask how much adding EFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFR grows to \$2676.66.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFR is available.

8 Conclusion

This study provides compelling evidence that Equity Financing Restraint (EFR) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EFR maintains its predictive power even after accounting for transaction costs and controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EFR generates substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.47 and statistically significant alphas relative to both the Fama-French five-factor model plus momentum (21 bps/month, t -statistic = 2.84) and an extended model including six closely related equity issuance and growth-based factors (15 bps/month, t -statistic = 2.24). The persistence of significant alpha even after controlling for conceptually similar strategies—including share issuance measures, book equity growth, net payout yield, and asset growth—suggests that EFR captures unique information about firm financing decisions and their implications for future returns that is not fully subsumed by existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The signal’s ability to generate economically meaningful returns after transaction costs suggests potential implementation value in quantitative investment strategies. Moreover, the distinct information content of EFR relative to other financing-based anomalies indicates that firms’ equity financing constraints and decisions contain nuanced information about future performance that warrants careful consideration in portfolio construction and risk management.

However, this study is subject to several limitations that should be acknowledged. First, while we employ state-of-the-art methodologies to account for data-snooping

concerns, the proliferation of factors in the literature raises ongoing questions about the true out-of-sample performance of any newly documented anomaly. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, while we document the statistical and economic significance of EFR, the underlying economic mechanisms driving this predictability deserve further theoretical and empirical investigation.

Future research should explore several promising avenues. First, examining the performance of EFR in international markets would provide valuable insights into the universality of this phenomenon. Second, investigating the time-varying nature of the EFR premium and its relationship to macroeconomic conditions, market sentiment, and regulatory changes could enhance our understanding of when and why this signal is most effective. Third, developing and testing economic theories that explain why equity financing restraint predicts returns would strengthen the theoretical foundation of this anomaly. Finally, exploring the interaction between EFR and corporate governance, information asymmetry, or behavioral biases may reveal important cross-sectional variations in the signal's effectiveness and provide deeper insights into the economic forces at play.

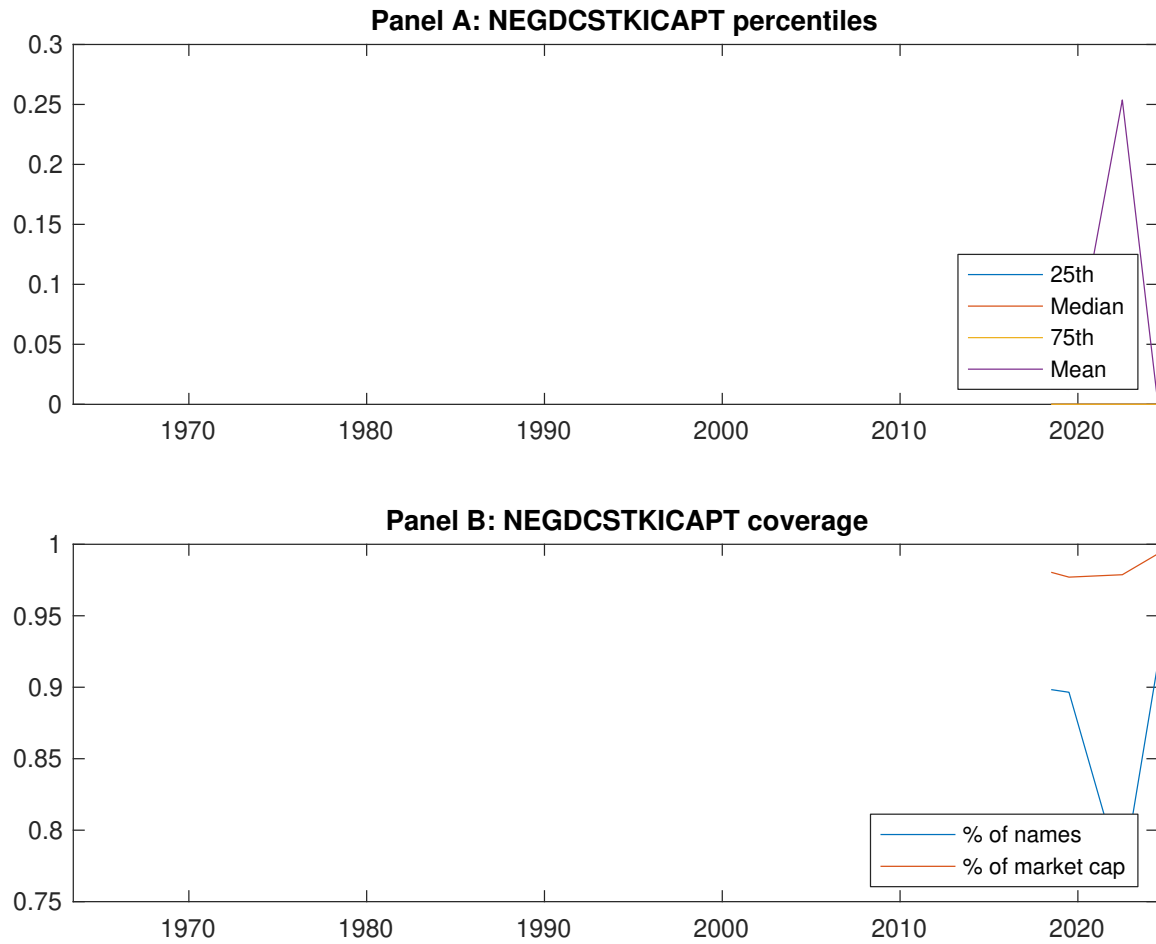


Figure 1: Times series of EFR percentiles and coverage.
This figure plots descriptive statistics for EFR. Panel A shows cross-sectional percentiles of EFR over the sample. Panel B plots the monthly coverage of EFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43 [2.56]	0.55 [3.05]	0.62 [3.47]	0.67 [4.10]	0.74 [4.68]	0.31 [4.15]
α_{CAPM}	-0.14 [-2.79]	-0.07 [-1.62]	0.01 [0.13]	0.11 [2.34]	0.20 [4.35]	0.34 [4.56]
α_{FF3}	-0.14 [-2.81]	-0.06 [-1.37]	0.02 [0.51]	0.07 [1.61]	0.16 [3.57]	0.30 [4.02]
α_{FF4}	-0.12 [-2.35]	-0.03 [-0.65]	0.04 [0.82]	0.03 [0.73]	0.15 [3.38]	0.27 [3.59]
α_{FF5}	-0.16 [-3.24]	0.00 [0.01]	0.04 [0.78]	0.01 [0.12]	0.07 [1.64]	0.24 [3.16]
α_{FF6}	-0.15 [-2.85]	0.02 [0.50]	0.05 [1.02]	-0.02 [-0.49]	0.07 [1.71]	0.22 [2.93]
Panel B: Fama and French (2018) 6-factor model loadings for EFR-sorted portfolios						
β_{MKT}	0.97 [78.87]	1.02 [94.38]	1.02 [88.97]	1.01 [91.17]	0.97 [93.02]	0.00 [0.01]
β_{SMB}	-0.02 [-0.86]	0.02 [1.42]	0.04 [2.53]	-0.07 [-4.55]	-0.01 [-0.42]	0.01 [0.34]
β_{HML}	0.05 [2.13]	-0.01 [-0.39]	-0.07 [-3.20]	0.08 [3.91]	0.02 [1.03]	-0.03 [-0.85]
β_{RMW}	0.13 [5.34]	-0.11 [-5.31]	-0.03 [-1.56]	0.10 [4.46]	0.13 [6.13]	-0.00 [-0.10]
β_{CMA}	-0.11 [-3.04]	-0.10 [-3.23]	0.01 [0.28]	0.15 [4.80]	0.24 [7.92]	0.34 [6.65]
β_{UMD}	-0.03 [-2.14]	-0.03 [-2.93]	-0.02 [-1.52]	0.04 [3.67]	-0.01 [-0.59]	0.02 [1.12]
Panel C: Average number of firms (n) and market capitalization (me)						
n	759	685	573	674	736	
me (\$10 ⁶)	1751	1481	2106	2276	2519	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.15]	0.34 [4.56]	0.30 [4.02]	0.27 [3.59]	0.24 [3.16]	0.22 [2.93]
Quintile	NYSE	EW	0.53 [8.77]	0.60 [10.30]	0.53 [9.67]	0.45 [8.39]	0.41 [7.82]	0.36 [6.87]
Quintile	Name	VW	0.31 [4.09]	0.33 [4.29]	0.30 [3.84]	0.27 [3.42]	0.24 [3.17]	0.23 [2.93]
Quintile	Cap	VW	0.24 [3.14]	0.26 [3.43]	0.23 [3.07]	0.19 [2.47]	0.21 [2.75]	0.18 [2.31]
Decile	NYSE	VW	0.26 [2.94]	0.28 [3.07]	0.21 [2.41]	0.17 [1.91]	0.21 [2.39]	0.18 [2.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.65]	0.30 [4.06]	0.26 [3.54]	0.25 [3.32]	0.22 [2.96]	0.21 [2.84]
Quintile	NYSE	EW	0.34 [5.00]	0.42 [6.44]	0.35 [5.70]	0.31 [5.12]	0.23 [4.05]	0.21 [3.67]
Quintile	Name	VW	0.27 [3.59]	0.29 [3.81]	0.25 [3.36]	0.24 [3.13]	0.22 [2.92]	0.21 [2.79]
Quintile	Cap	VW	0.20 [2.66]	0.22 [2.96]	0.19 [2.59]	0.17 [2.26]	0.18 [2.47]	0.17 [2.24]
Decile	NYSE	VW	0.22 [2.46]	0.23 [2.59]	0.17 [1.98]	0.15 [1.71]	0.18 [2.01]	0.16 [1.83]

Table 3: Conditional sort on size and EFR

This table presents results for conditional double sorts on size and EFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFR and short stocks with low EFR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EFR Quintiles					EFR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.48 [1.92]	0.65 [2.57]	0.87 [3.46]	0.89 [3.62]	1.02 [4.46]	0.56 [7.07]	0.61 [7.75]	0.56 [7.33]	0.51 [6.56]	0.45 [5.92]	0.42 [5.38]
	(2)	0.56 [2.46]	0.71 [3.01]	0.90 [3.82]	0.89 [3.99]	0.96 [4.49]	0.41 [4.81]	0.46 [5.40]	0.38 [4.62]	0.33 [3.95]	0.32 [3.90]	0.29 [3.43]
	(3)	0.56 [2.78]	0.64 [3.00]	0.76 [3.46]	0.84 [4.14]	0.91 [4.70]	0.35 [4.76]	0.38 [5.04]	0.33 [4.45]	0.29 [3.87]	0.29 [3.86]	0.26 [3.44]
	(4)	0.51 [2.68]	0.61 [3.10]	0.78 [3.84]	0.77 [4.12]	0.81 [4.50]	0.30 [4.03]	0.34 [4.53]	0.27 [3.83]	0.26 [3.59]	0.13 [1.84]	0.13 [1.86]
	(5)	0.45 [2.74]	0.48 [2.66]	0.50 [2.89]	0.56 [3.41]	0.69 [4.36]	0.24 [2.68]	0.25 [2.73]	0.22 [2.41]	0.18 [1.93]	0.21 [2.28]	0.18 [1.93]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EFR Quintiles					EFR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	382	383	382	381	374	31	33	41	30	29	
	(2)	109	107	108	108	106	57	56	58	56	56	
	(3)	78	78	77	77	78	98	95	98	100	100	
	(4)	65	65	65	65	66	204	205	216	212	218	
(5)	60	60	60	60	60	1478	1429	1664	1677	1898		

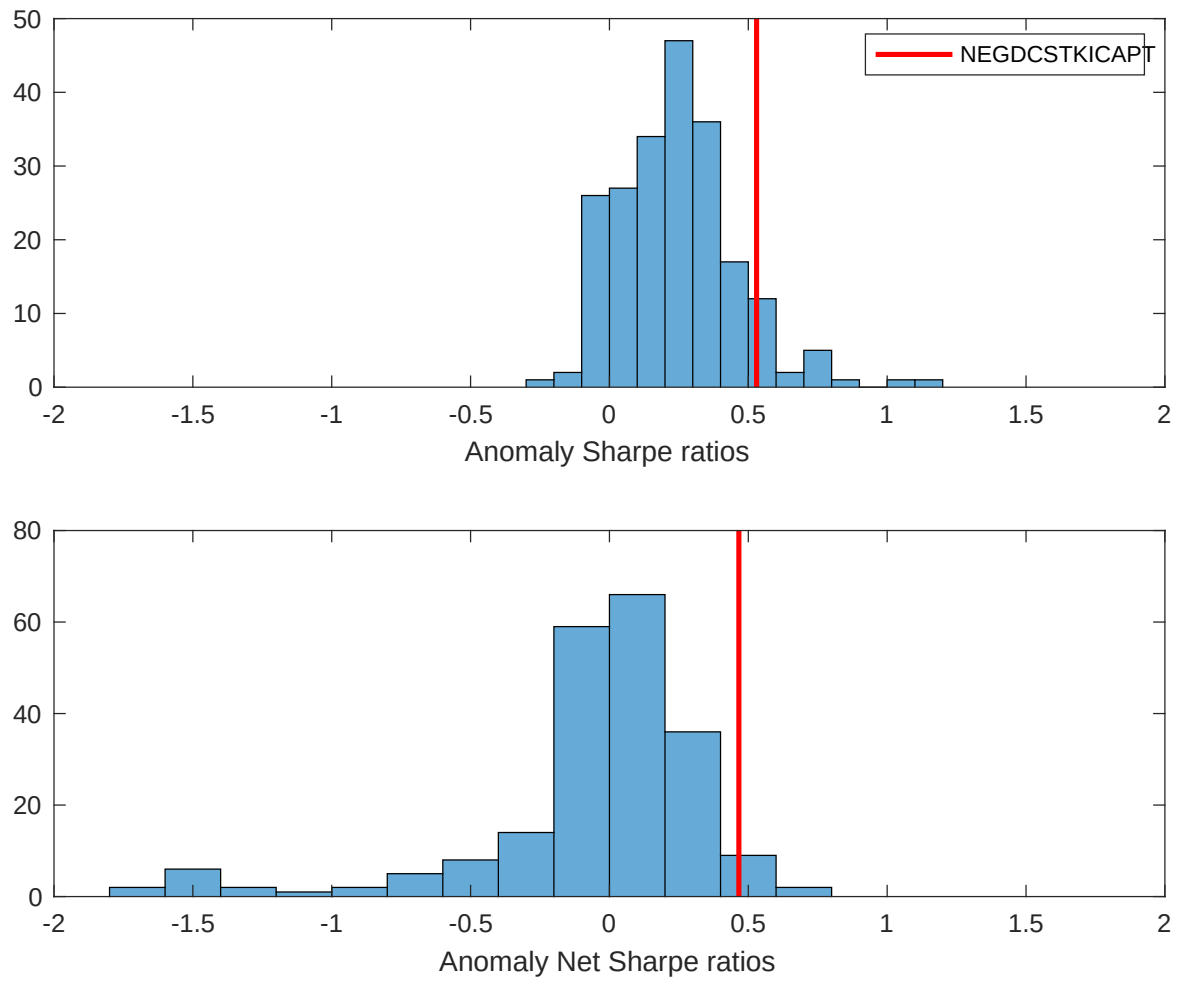


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

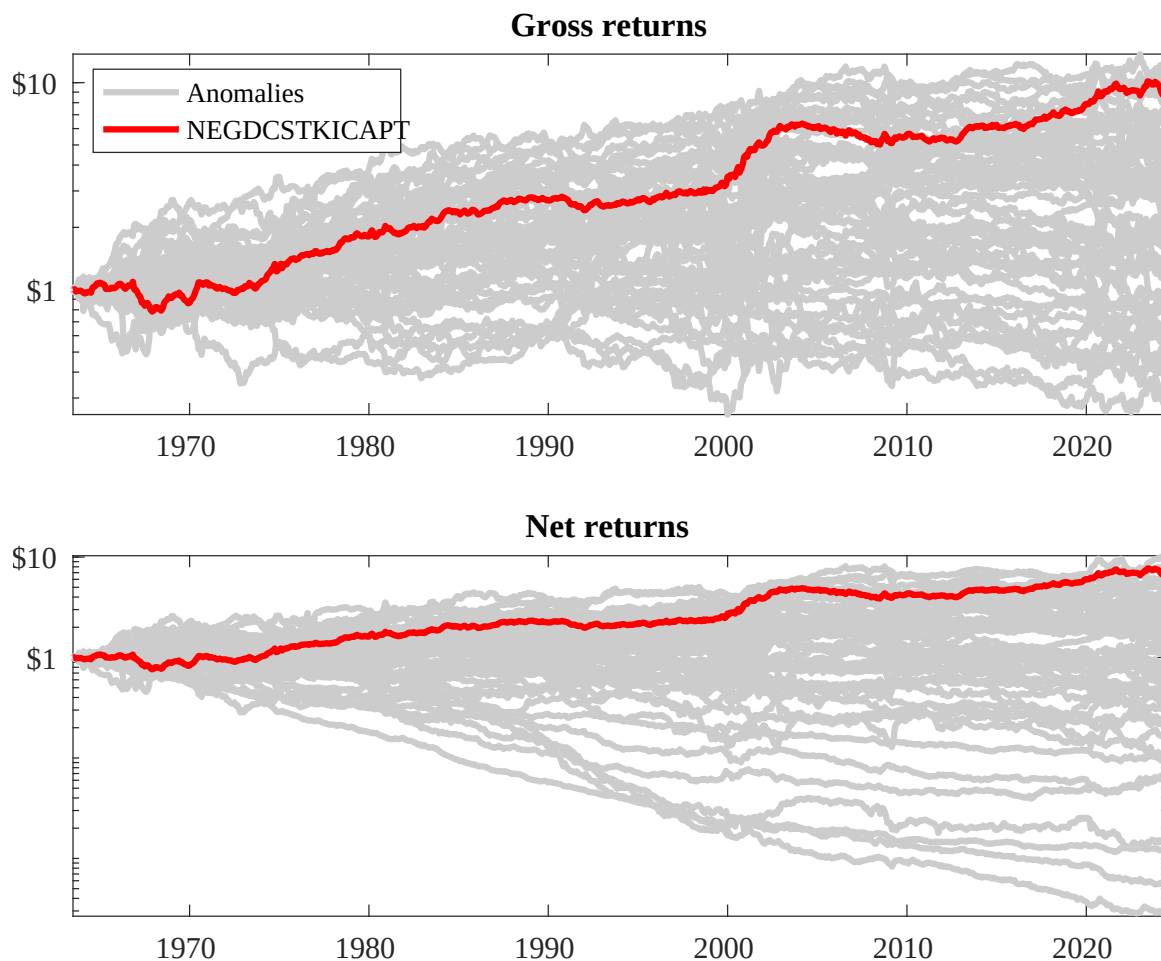
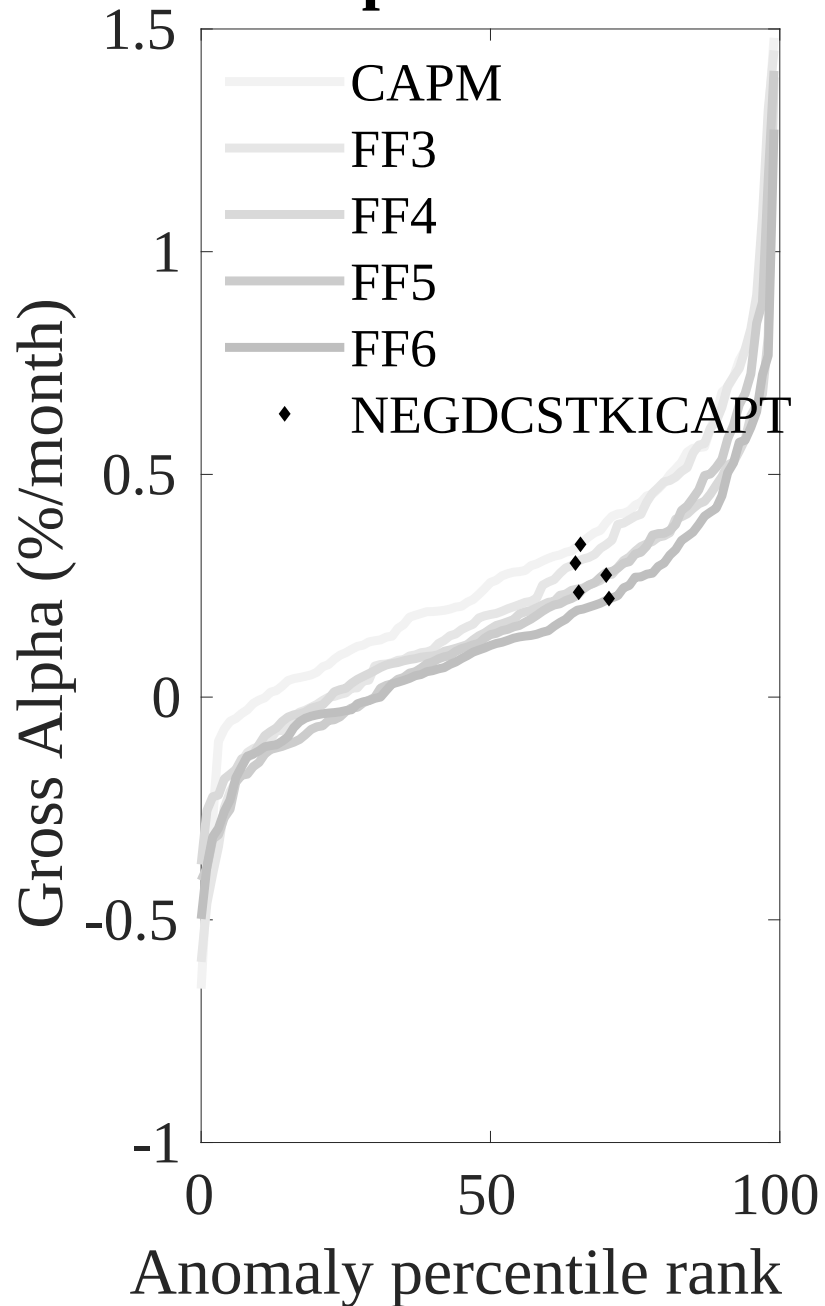


Figure 3: Dollar invested.

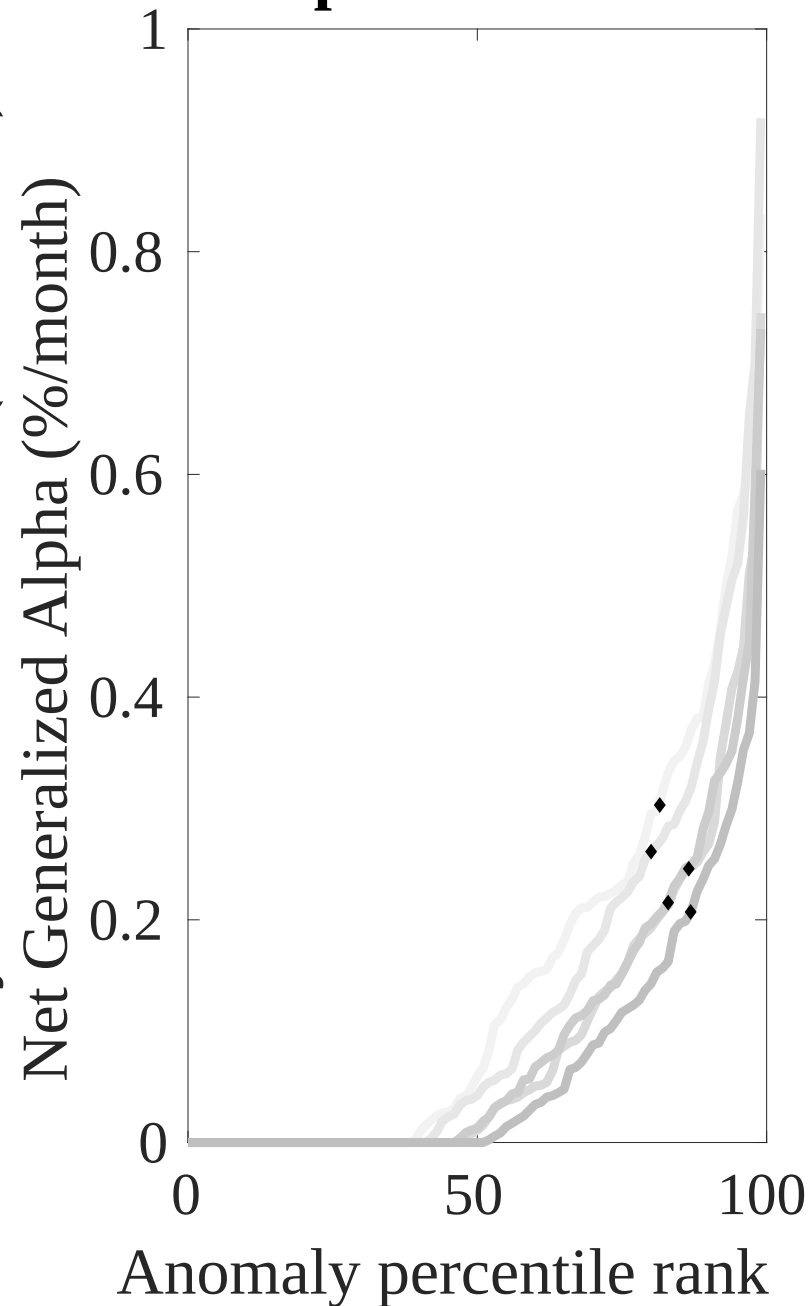
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



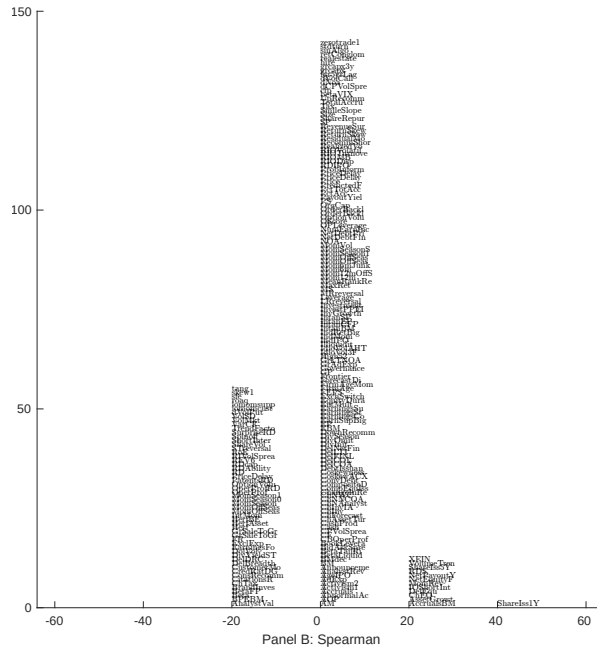
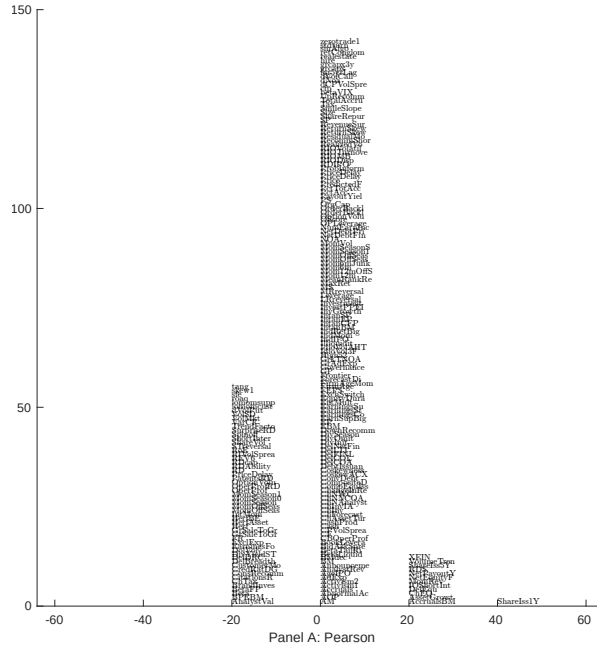


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

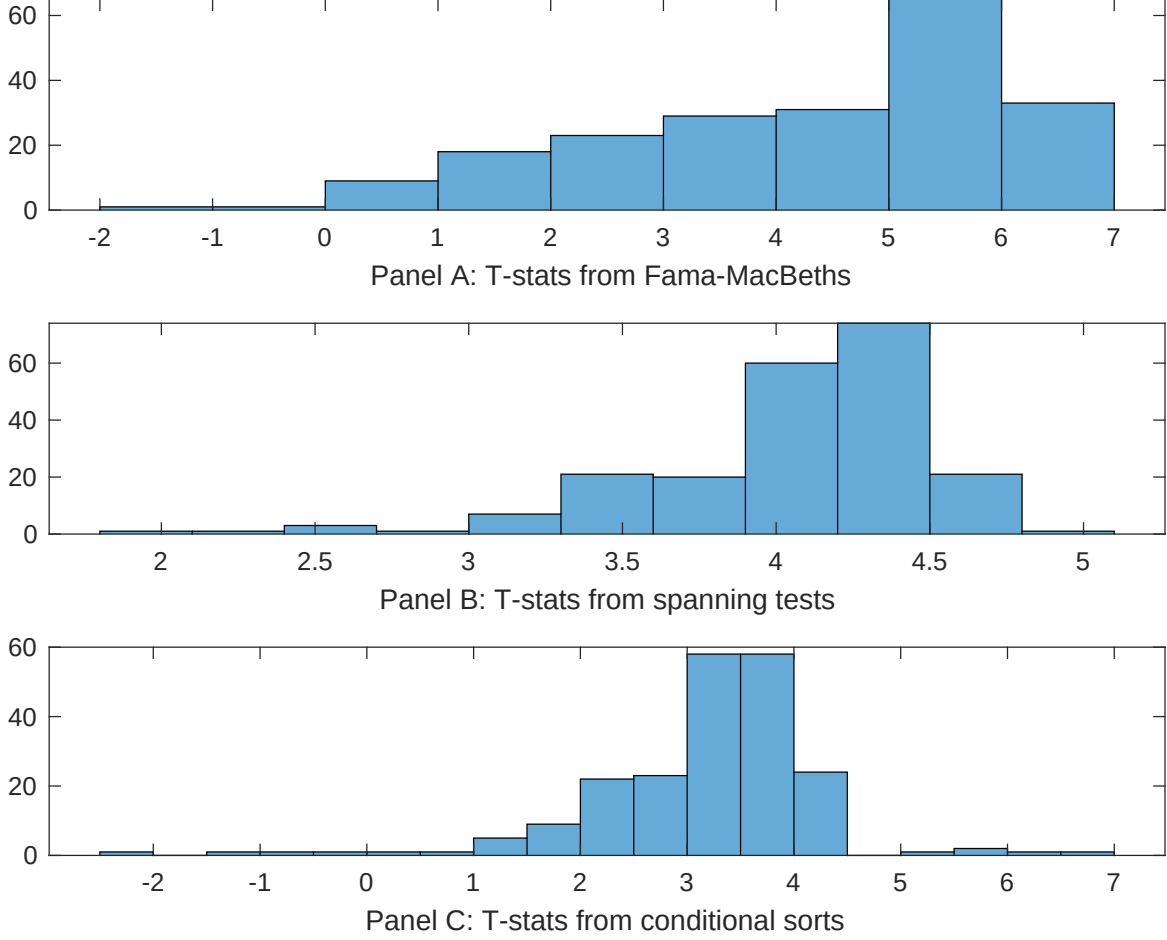


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.07]	0.16 [6.82]	0.13 [6.45]	0.12 [5.43]	0.12 [5.71]	0.13 [6.17]	0.12 [3.91]
EFR	0.40 [4.82]	0.36 [3.88]	0.36 [4.39]	0.30 [2.83]	0.44 [4.82]	0.37 [3.99]	0.15 [1.47]
Anomaly 1	0.13 [5.87]						0.66 [2.40]
Anomaly 2		0.37 [3.17]					-0.13 [-0.59]
Anomaly 3			0.26 [4.77]				0.24 [4.27]
Anomaly 4				0.21 [1.85]			0.93 [0.88]
Anomaly 5					0.11 [3.08]		-0.64 [-0.11]
Anomaly 6						0.92 [7.86]	0.65 [5.45]
# months	715	720	715	715	720	720	715
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.76]	0.20 [2.83]	0.15 [1.99]	0.25 [3.36]	0.22 [2.94]	0.21 [2.85]	0.15 [2.24]
Anomaly 1	30.08 [8.25]						15.75 [3.61]
Anomaly 2		31.95 [8.21]					37.03 [6.67]
Anomaly 3			28.73 [7.40]				12.97 [3.00]
Anomaly 4				17.51 [6.16]			5.57 [1.83]
Anomaly 5					13.68 [3.70]		-9.63 [-1.89]
Anomaly 6						-3.55 [-0.80]	-21.73 [-4.55]
mkt	0.93 [0.54]	1.25 [0.73]	1.48 [0.85]	2.91 [1.62]	-0.05 [-0.03]	0.20 [0.11]	3.33 [1.96]
smb	3.19 [1.29]	0.93 [0.38]	3.91 [1.55]	4.84 [1.88]	1.83 [0.71]	2.29 [0.88]	5.15 [2.10]
hml	-2.68 [-0.81]	-5.72 [-1.71]	0.44 [0.13]	-8.67 [-2.42]	-3.59 [-1.04]	-1.61 [-0.46]	-6.15 [-1.82]
rmw	-16.68 [-4.29]	1.18 [0.35]	-10.68 [-2.92]	-10.58 [-2.79]	0.92 [0.26]	-0.28 [-0.08]	-15.74 [-3.90]
cma	14.90 [2.79]	2.53 [0.41]	19.63 [3.75]	18.81 [3.43]	19.71 [3.15]	37.77 [5.09]	13.12 [1.80]
umd	0.81 [0.47]	1.62 [0.95]	0.30 [0.18]	3.58 [2.05]	2.52 [1.43]	2.11 [1.18]	-0.51 [-0.30]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	18	17	16	14	11	10	26

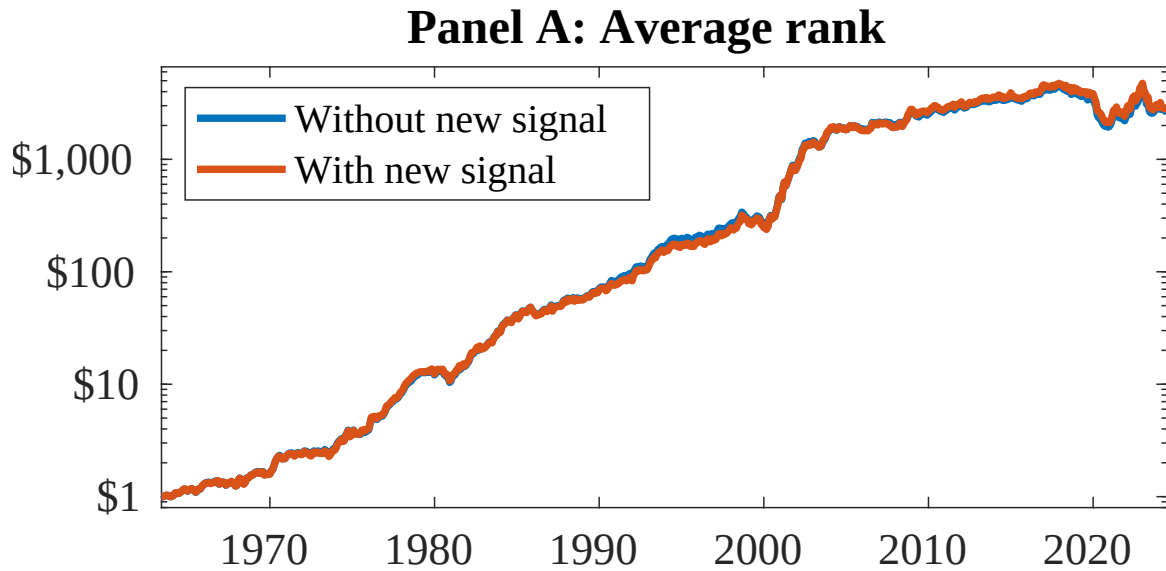


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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